



CLOUD COMPUTING

LAB 01

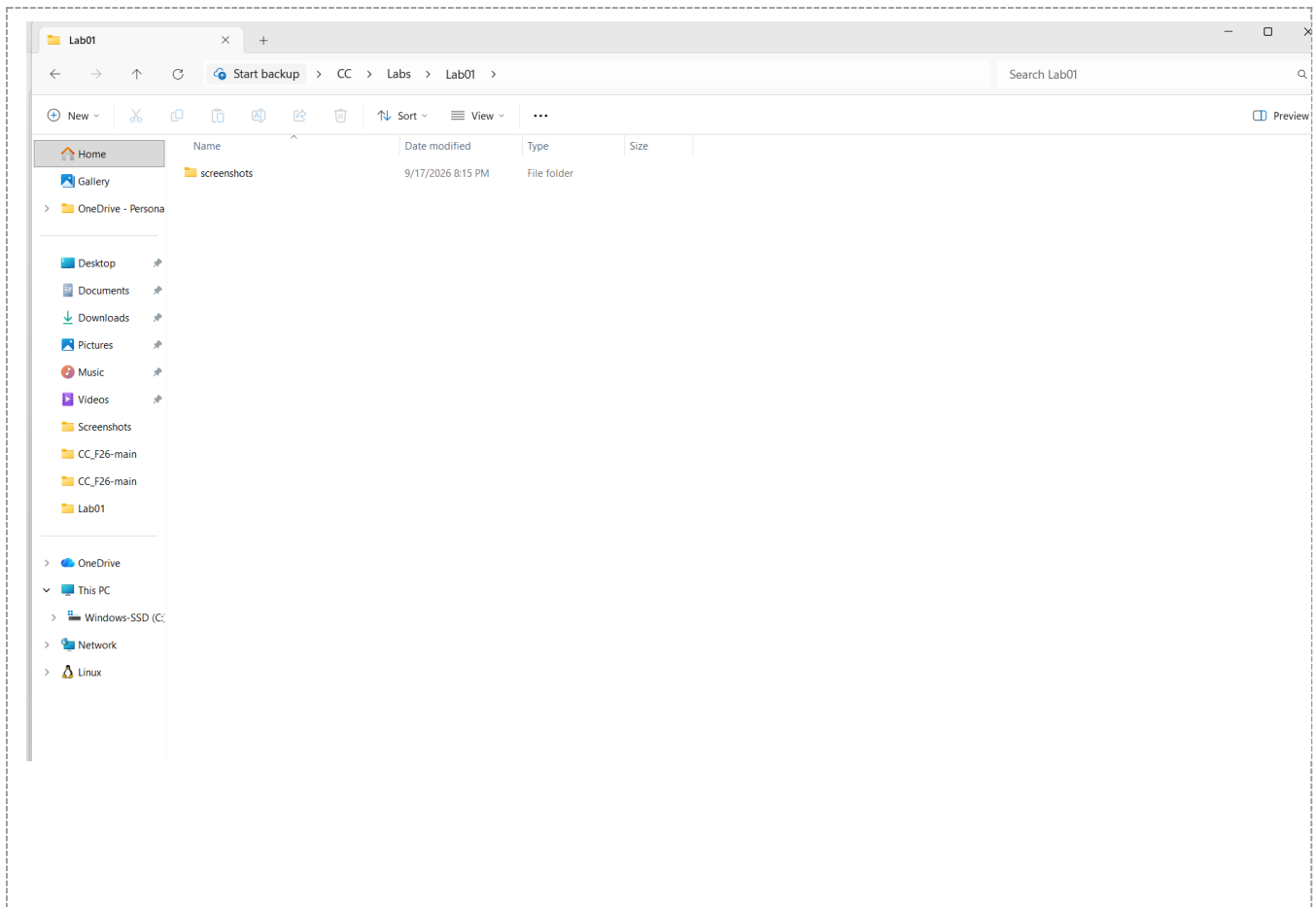
GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

- **Name:** Alizeh Anwar
- **Reg. no.:** FA24b1-SE-038
- **Course Code:** Cloud Computing (SE-665)
- **Section:** B
- **GitHub username:** alizeh-anwar

Getting Started

Create the directory Labs/Lab01/screenshots/ inside the CC repository.

lab01_folder_structure.png



Caption for lab01_folder_structure.png: Created the required Labs/Lab01/screenshots/ folder structure inside the repository.

Task 1: Create and Configure Your GitHub Account

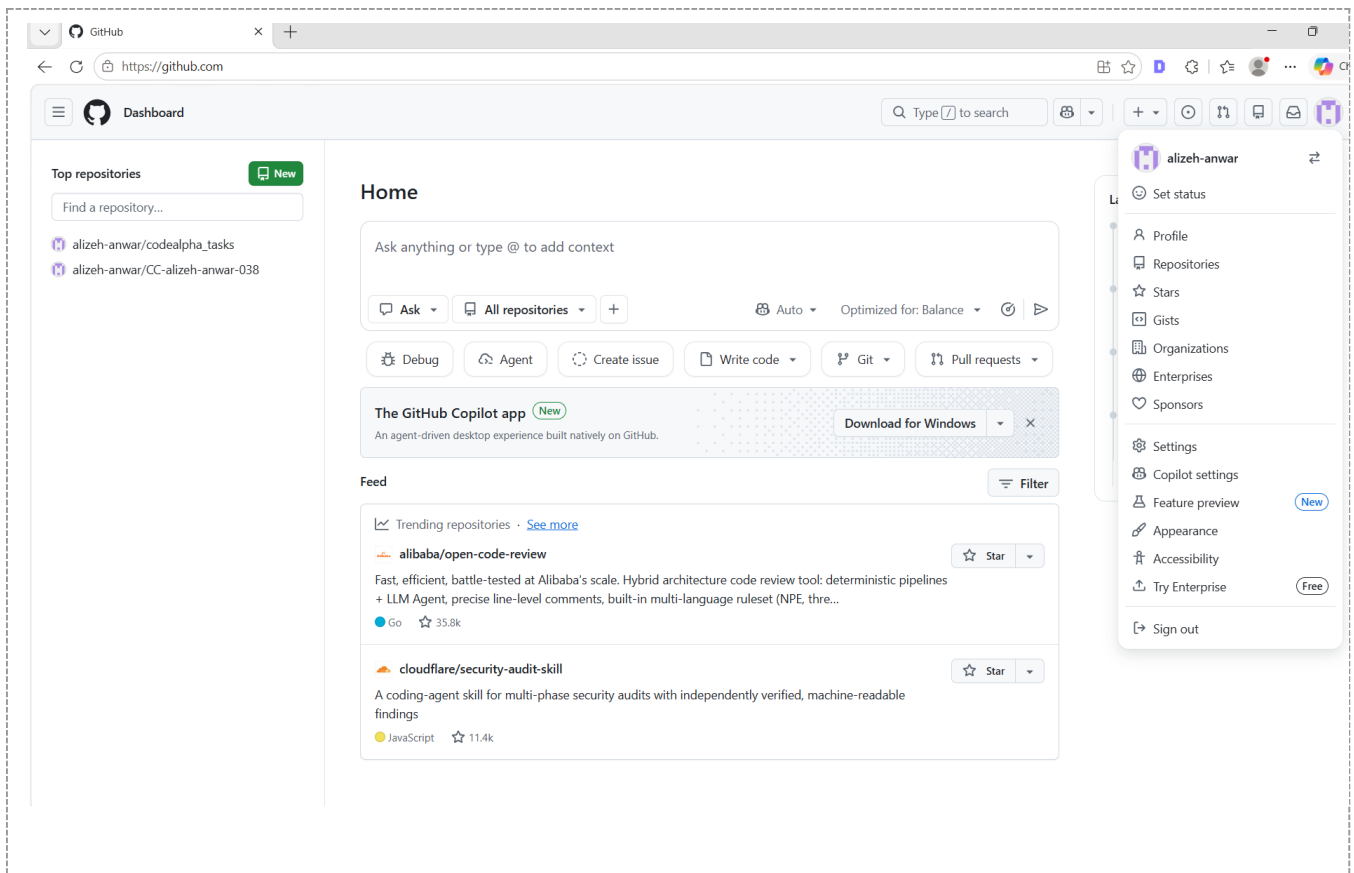
Create a personal GitHub account using your own email address and actual identity.

Choose a professional GitHub username related to your actual name (no spaces, prefer lowercase letters and numbers).

Go to GitHub > Settings > Public profile and enter your complete actual name in the Name field.

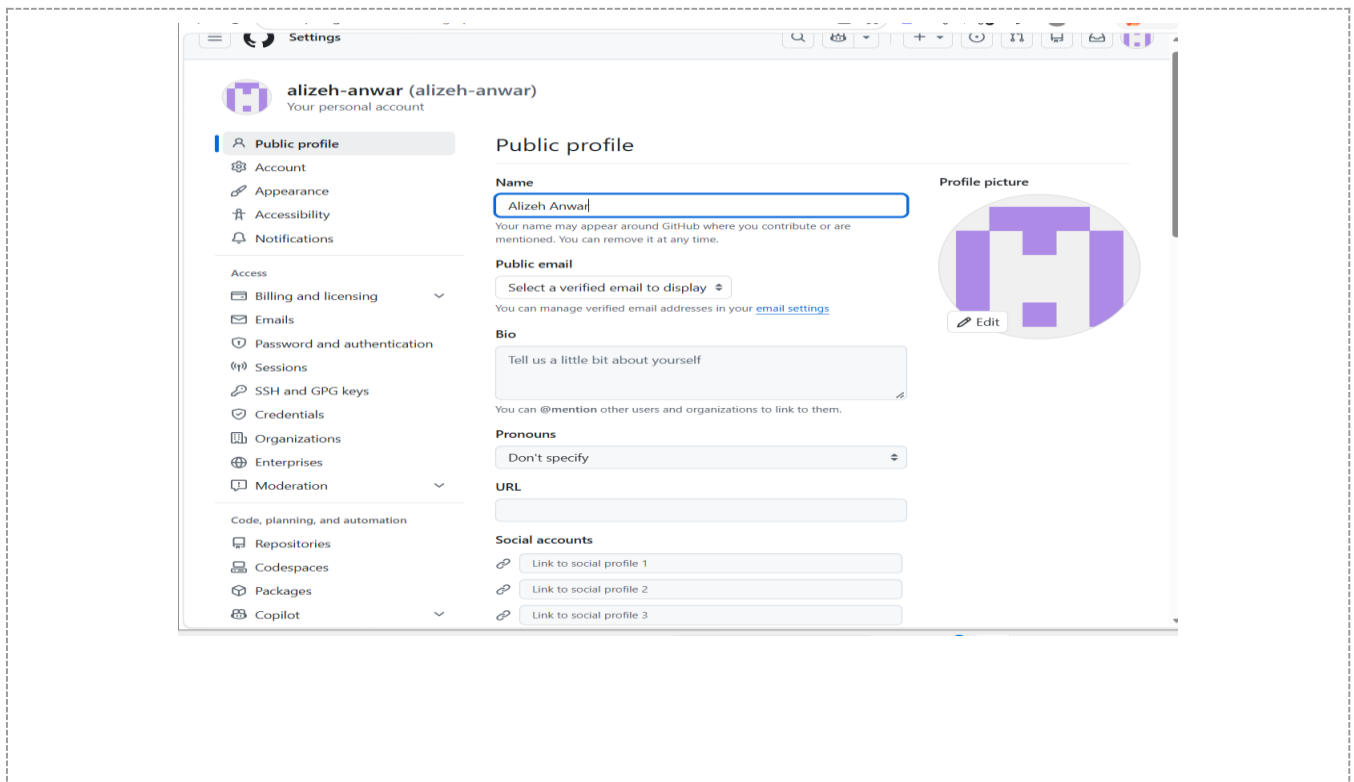
Do not include passwords, access tokens, or recovery codes in any screenshot.

github_profile.png



Caption for github_profile.png: GitHub profile overview showing the professional username alizeh-anwar.

github_actual_name.png



Caption for github_actual_name.png: GitHub public profile settings displaying the full actual name "Alizeh Anwar".

Task 2: Register in the Student Marks Portal

Open the Student Marks Portal and sign in using 'Continue with GitHub' with the account from Task 1.

Enter your actual registration number, correct course code, and correct section (A or B).

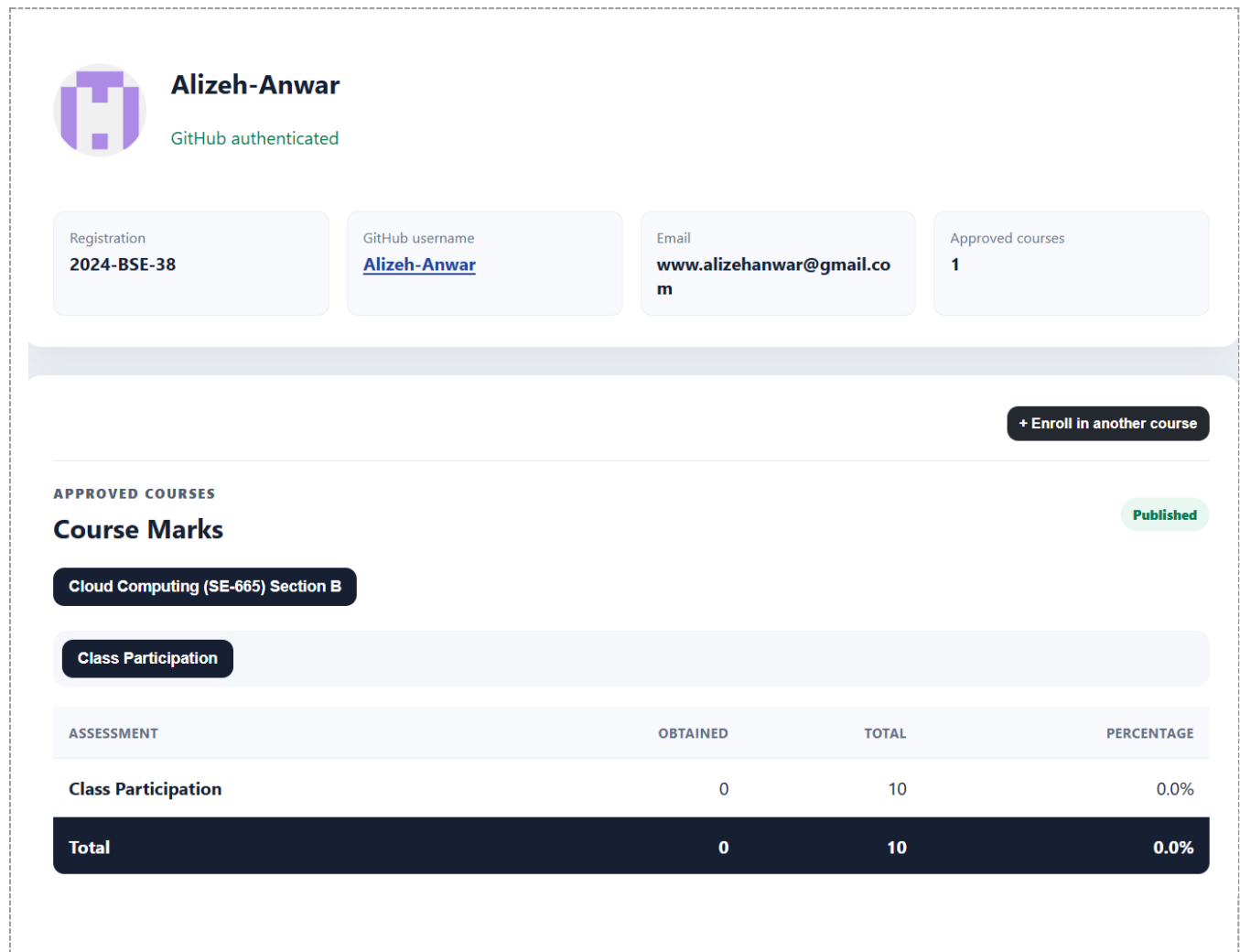
Review your information and submit the enrollment for approval.

portal_login_page.png



Caption for portal_login_page.png: Student Marks Portal login page prior to authenticating via GitHub.

portal_github_profile.png



Caption for portal_github_profile.png: Successful GitHub OAuth authentication on the Student Marks Portal.

portal_enrollment_submitted.png

+ Enroll in another course

APPROVED COURSES

Course Marks

Published

Cloud Computing (SE-665) Section B

Class Participation

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

Caption for portal_enrollment_submitted.png: Submitted course enrollment details (Reg No: FA24b1-SE-038, Course: SE-665, Section B).

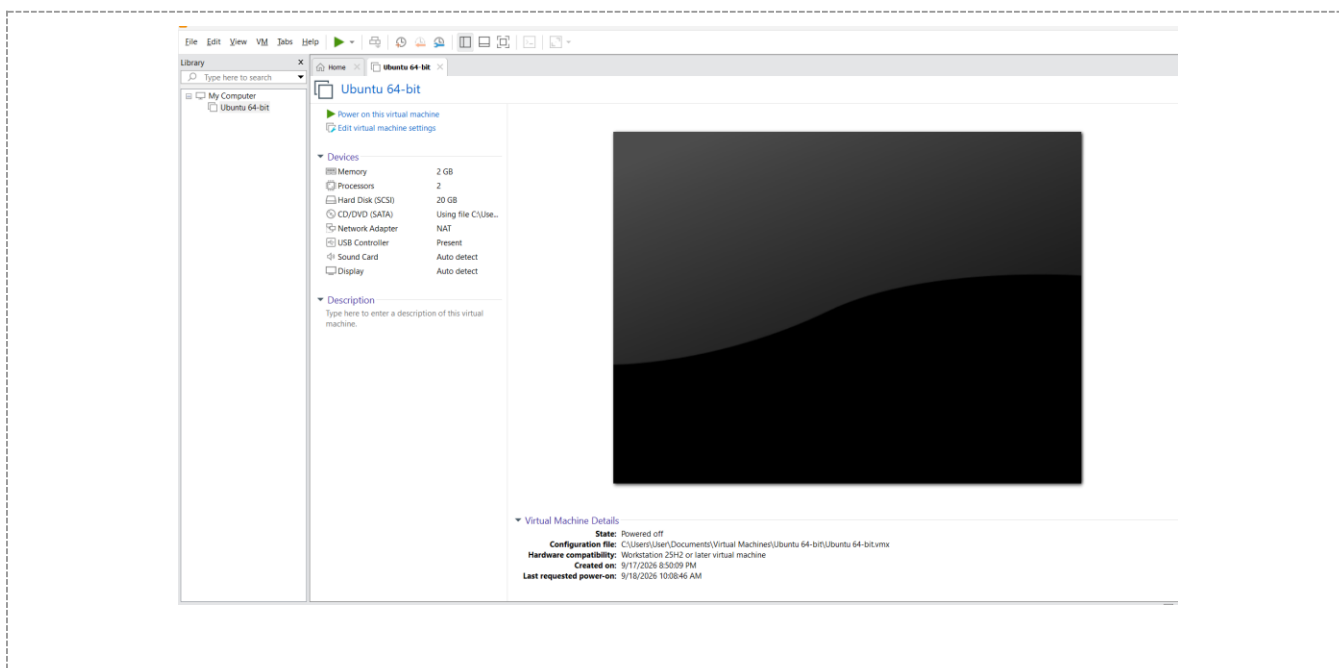
Task 3: Download VMware Workstation Pro and Ubuntu Server

Download and install VMware Workstation Pro; verify it opens and runs correctly.

Download the latest Ubuntu Server LTS ISO from the official Ubuntu download page.

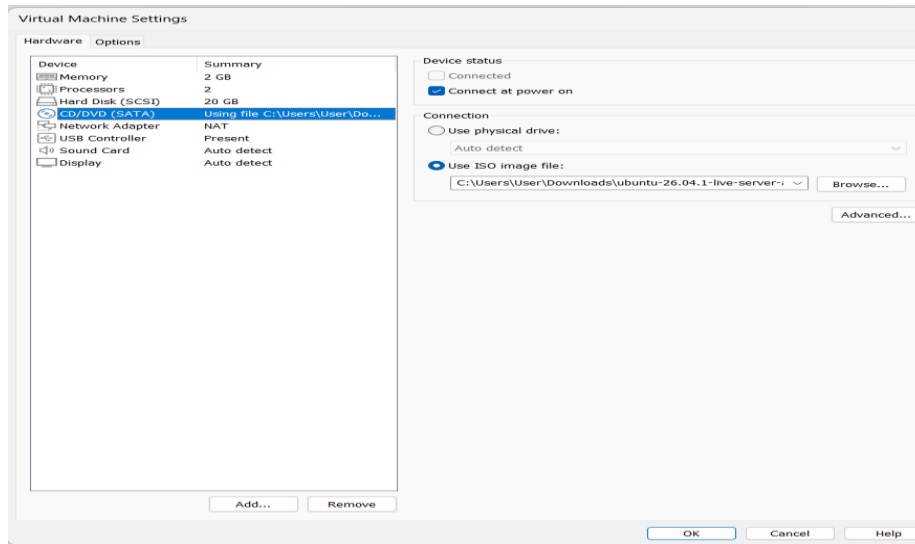
Confirm at least 20 GB of free storage is available for the virtual machine.

vmware_workstation.png



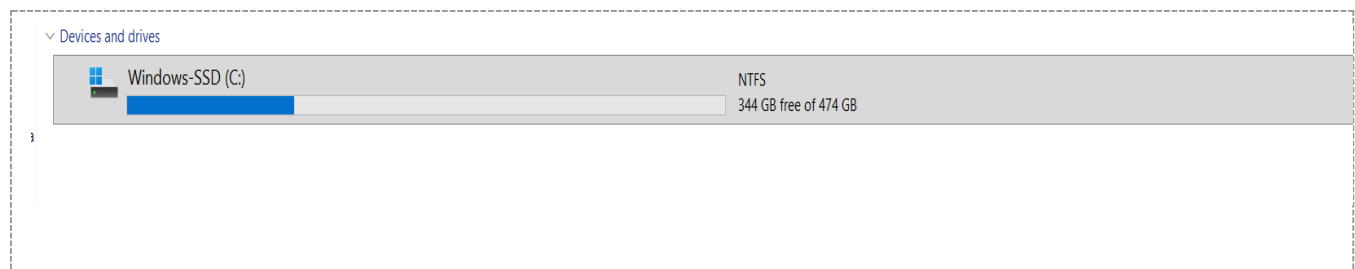
Caption for vmware_workstation.png: VMware Workstation Pro installed and running on the host machine.

ubuntu_server_iso.png



Caption for ubuntu_server_iso.png: Official Ubuntu Server LTS ISO image file downloaded successfully.

available_storage.png



Caption for available_storage.png: Host disk storage check confirming more than 20 GB free space.

Task 4: Create the Ubuntu Server Virtual Machine

Open VMware Workstation Pro and select 'Create a New Virtual Machine'.

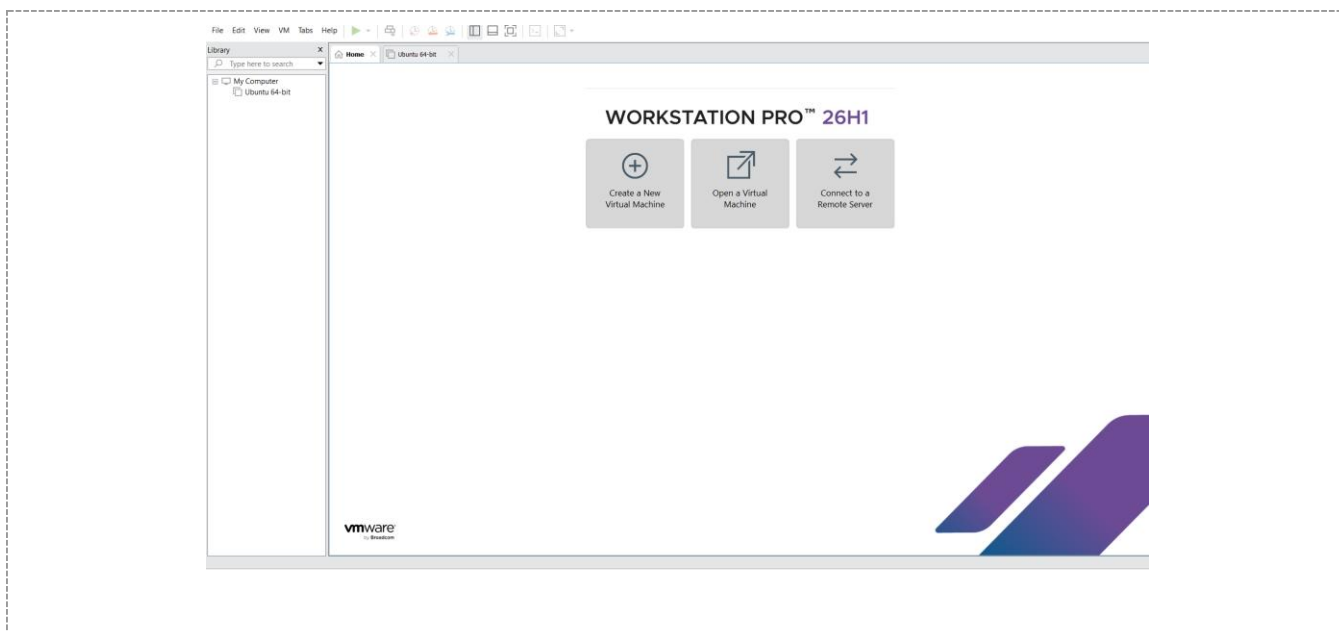
Select 'Typical (recommended)' configuration.

Select 'Installer disc image file (ISO)' and browse to the downloaded Ubuntu Server ISO.

Allocate at least 20 GB of virtual disk space and complete the wizard.

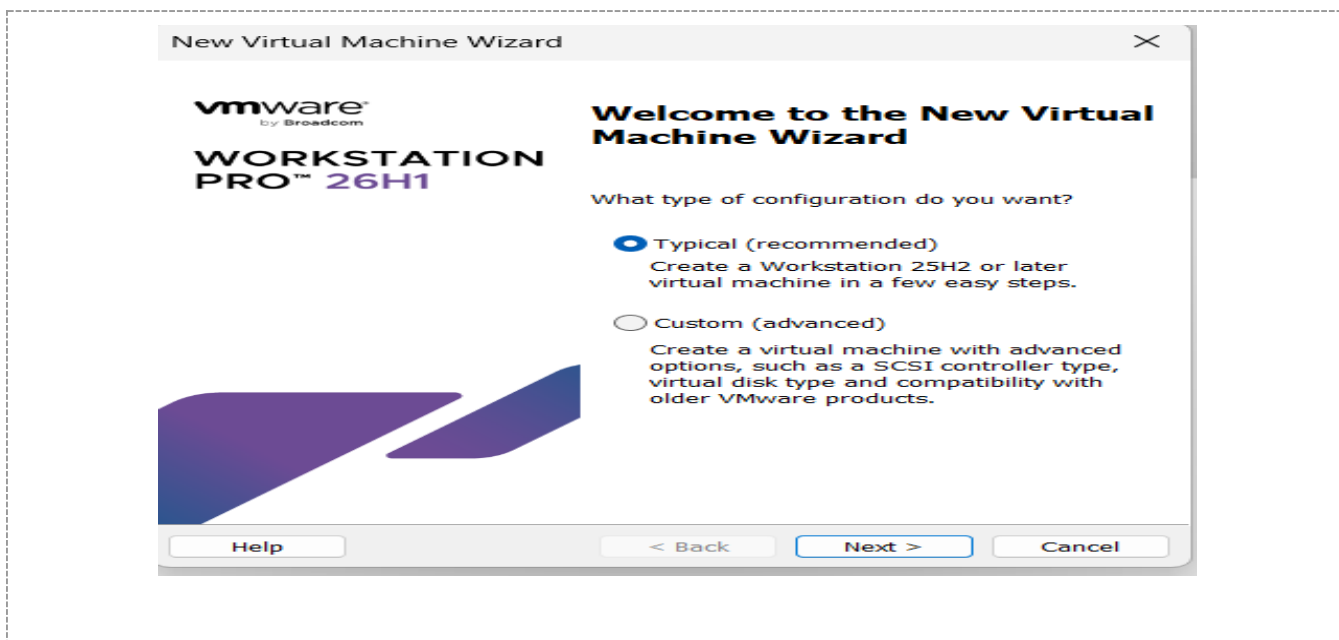
Power on the VM and wait for the Ubuntu Server installer to start.

vm_creation_wizard.png



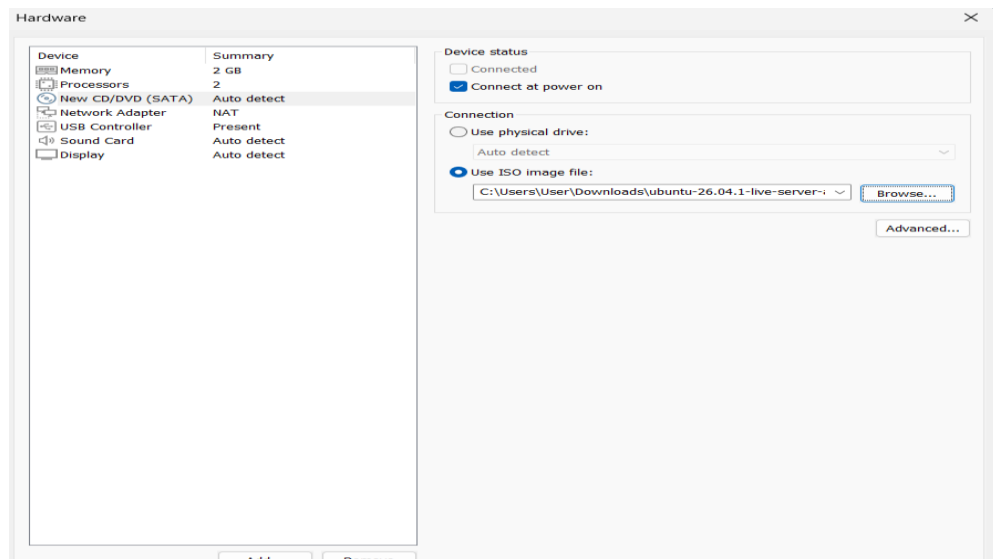
Caption for vm_creation_wizard.png: Launching the VMware New Virtual Machine Wizard.

vm_typical_configuration.png



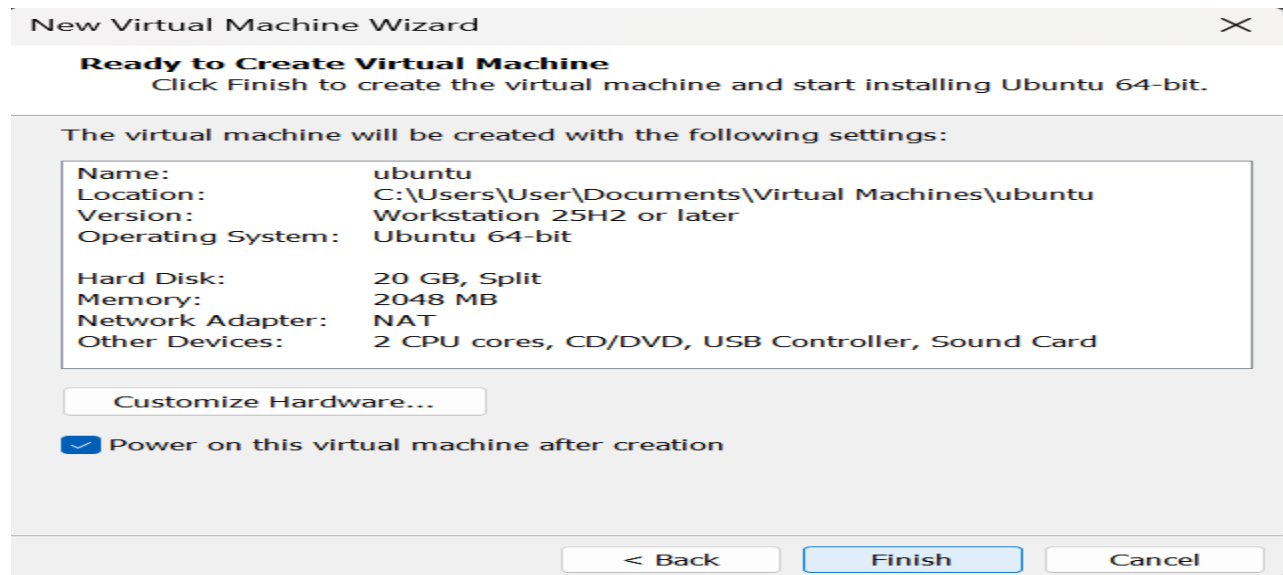
Caption for vm_typical_configuration.png: Selecting the Typical (recommended) VM configuration option.

vm_ubuntu_iso_selected.png



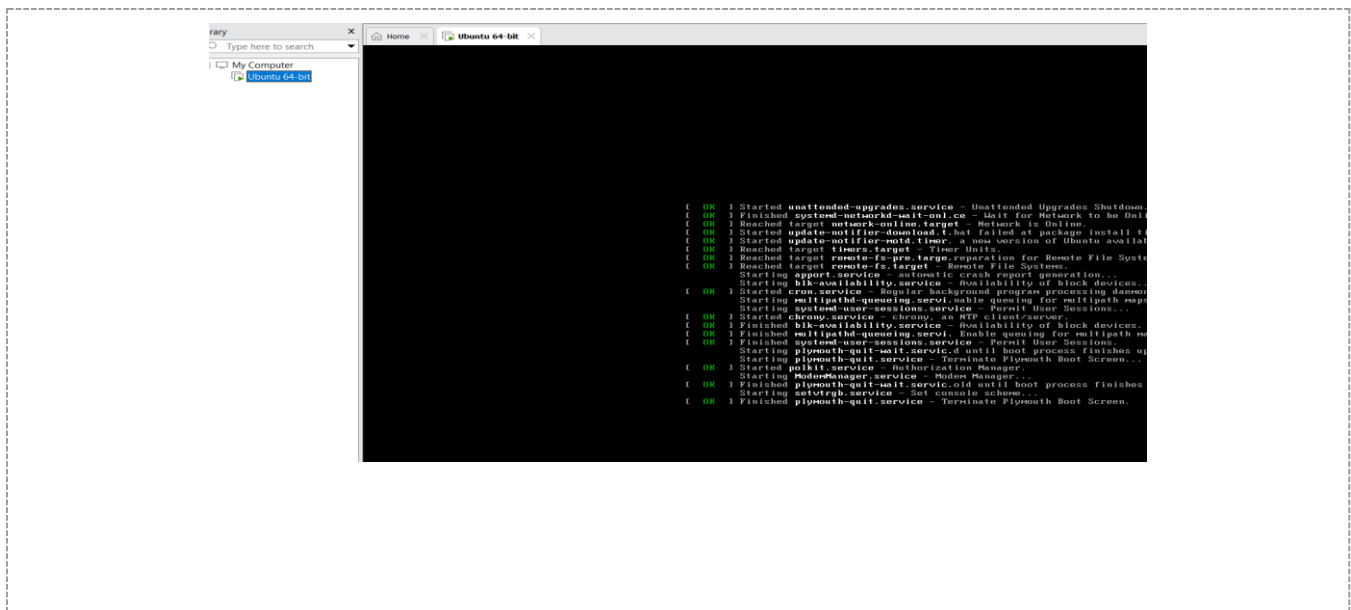
Caption for vm_ubuntu_iso_selected.png: Browsing and attaching the downloaded Ubuntu Server ISO file.

vm_configuration_summary.png



Caption for vm_configuration_summary.png: Reviewing final hardware and virtual disk settings (20 GB allocation).

ubuntu_installer_boot.png

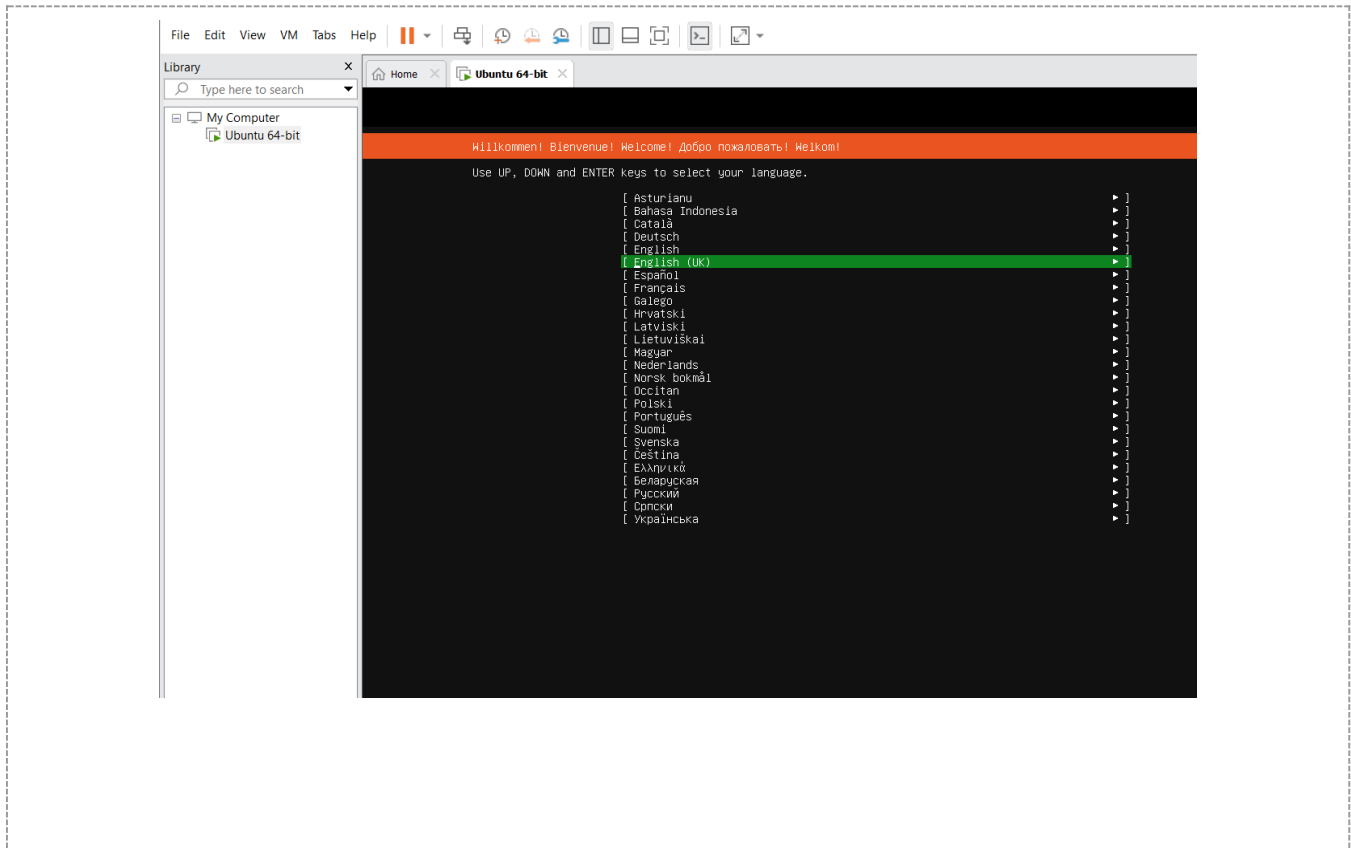


Caption for ubuntu_installer_boot.png: Powering on the virtual machine and booting into the Ubuntu Server installer menu.

Task 5: Choose the Installation Language

Use the arrow keys to select the required installation language and press Enter.

ubuntu_language.png



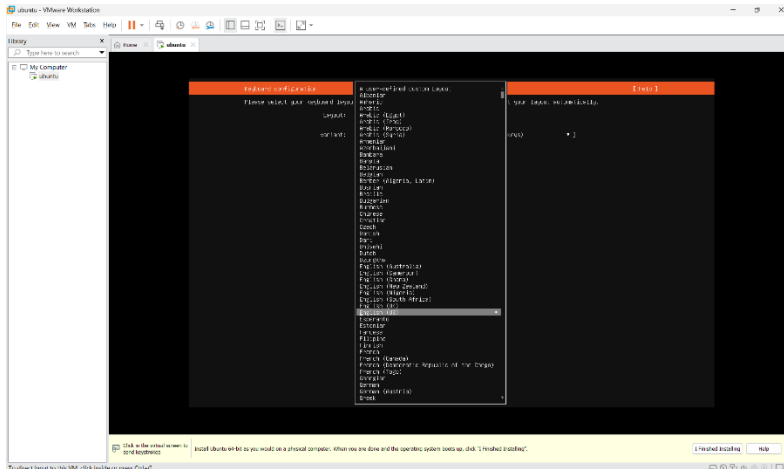
Caption for ubuntu_language.png: Selecting English as the primary installation language.

Task 6: Configure the Keyboard Layout

Select the keyboard layout that matches your keyboard.

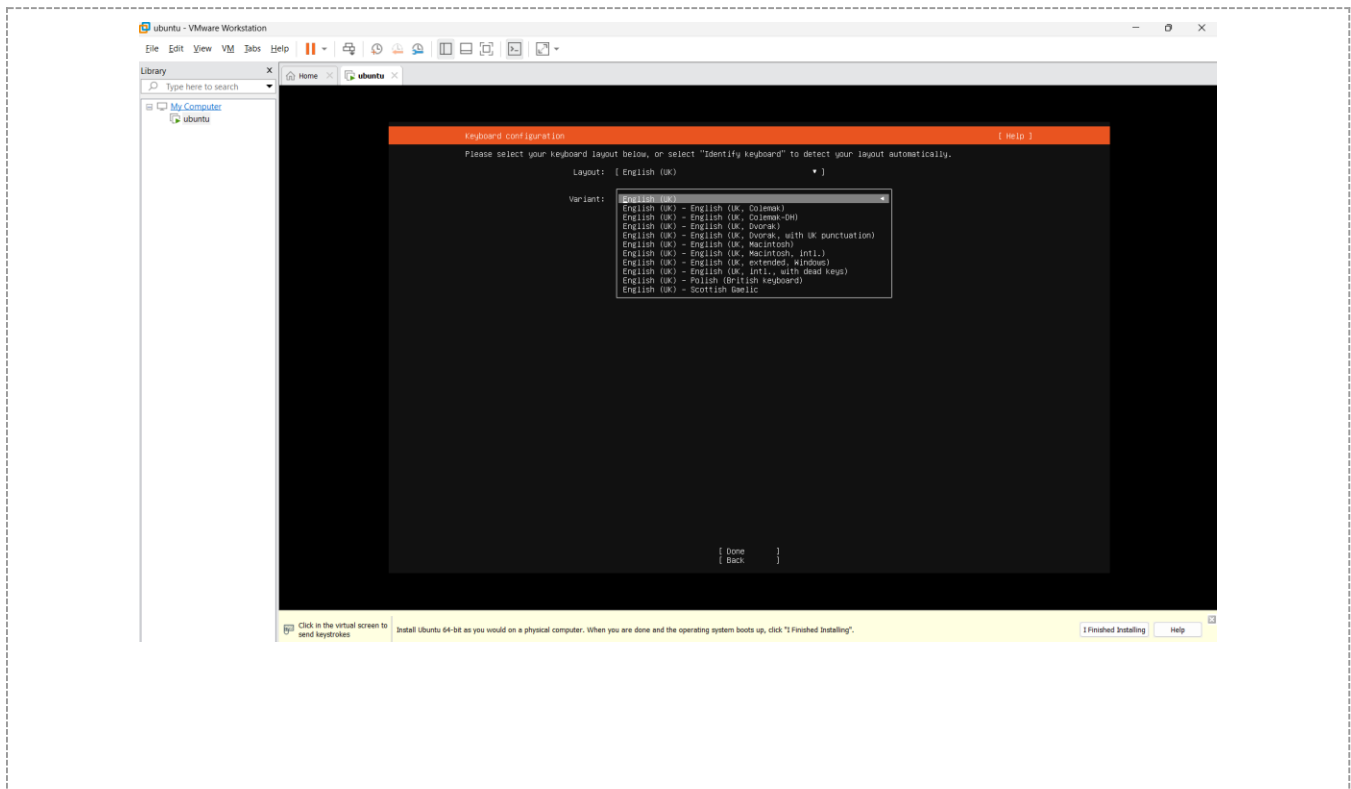
Select the appropriate keyboard variant, then select Done.

ubuntu_keyboard_layout.png



Caption for ubuntu_keyboard_layout.png: Choosing the default English (US) keyboard layout.

ubuntu_keyboard_variant.png

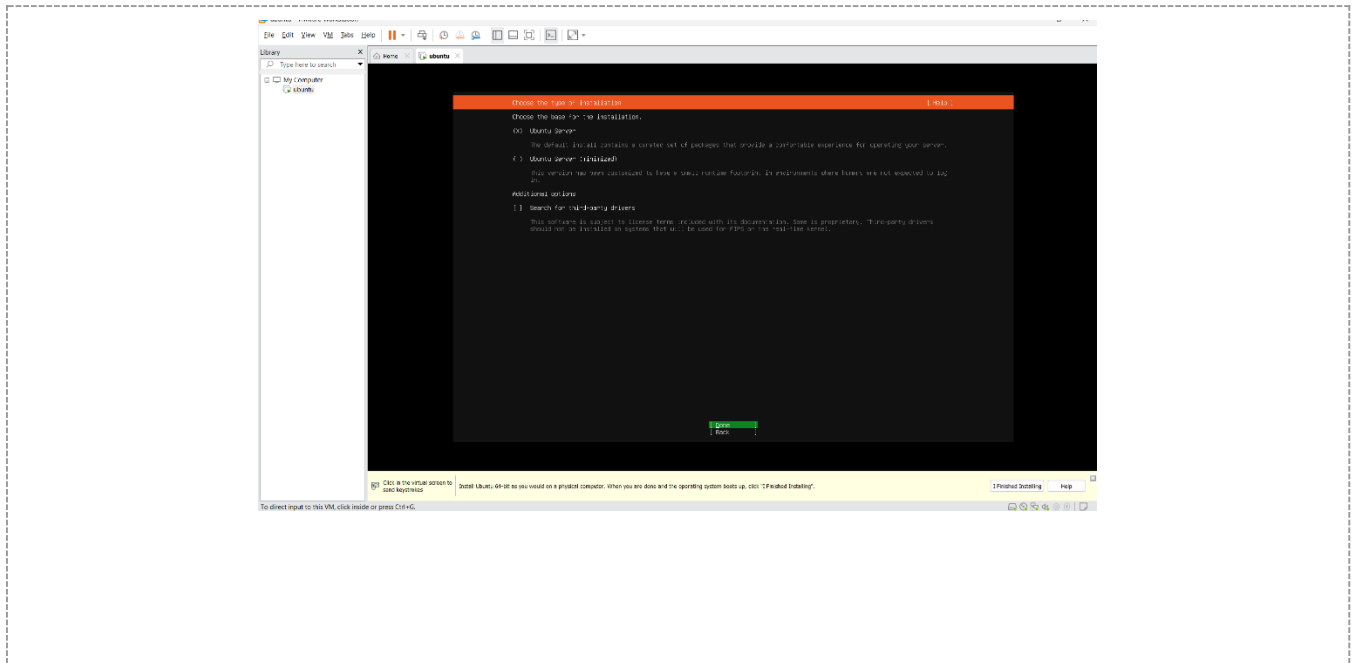


Caption for ubuntu_keyboard_variant.png: Confirming the standard keyboard variant selection.

Task 7: Choose the Ubuntu Server Installation Type

Select the normal Ubuntu Server installation option (not MAAS) and continue.

ubuntu_installation_type.png



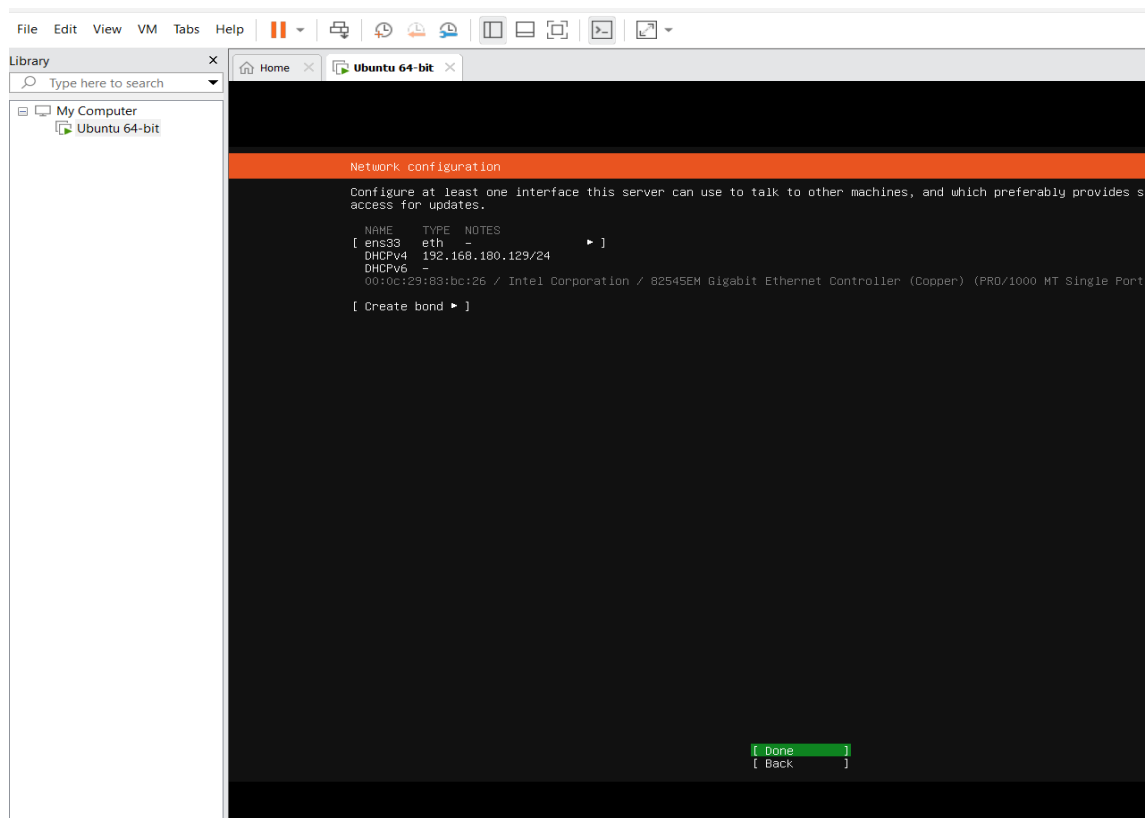
Caption for ubuntu_installation_type.png: Selecting the default base Ubuntu Server installation profile.

Task 8: Configure the Network

Review the network interface detected by the installer.

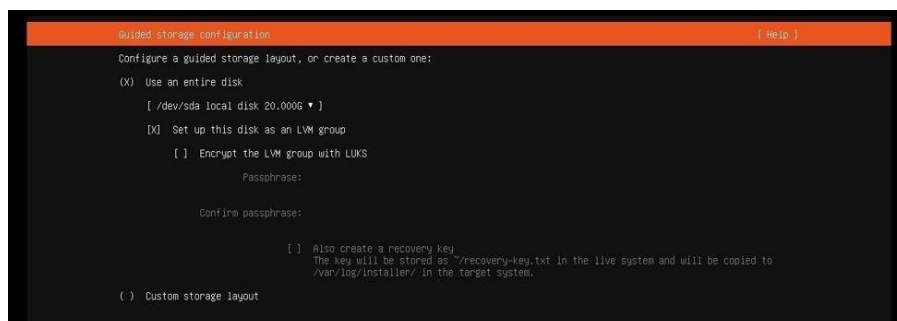
Confirm the interface received an IP address automatically via DHCP, then continue.

ubuntu_network_interface.png



Caption for `ubuntu_network_interface.png`: Network interface detection screen showing active connection.

ubuntu_installer_network.png



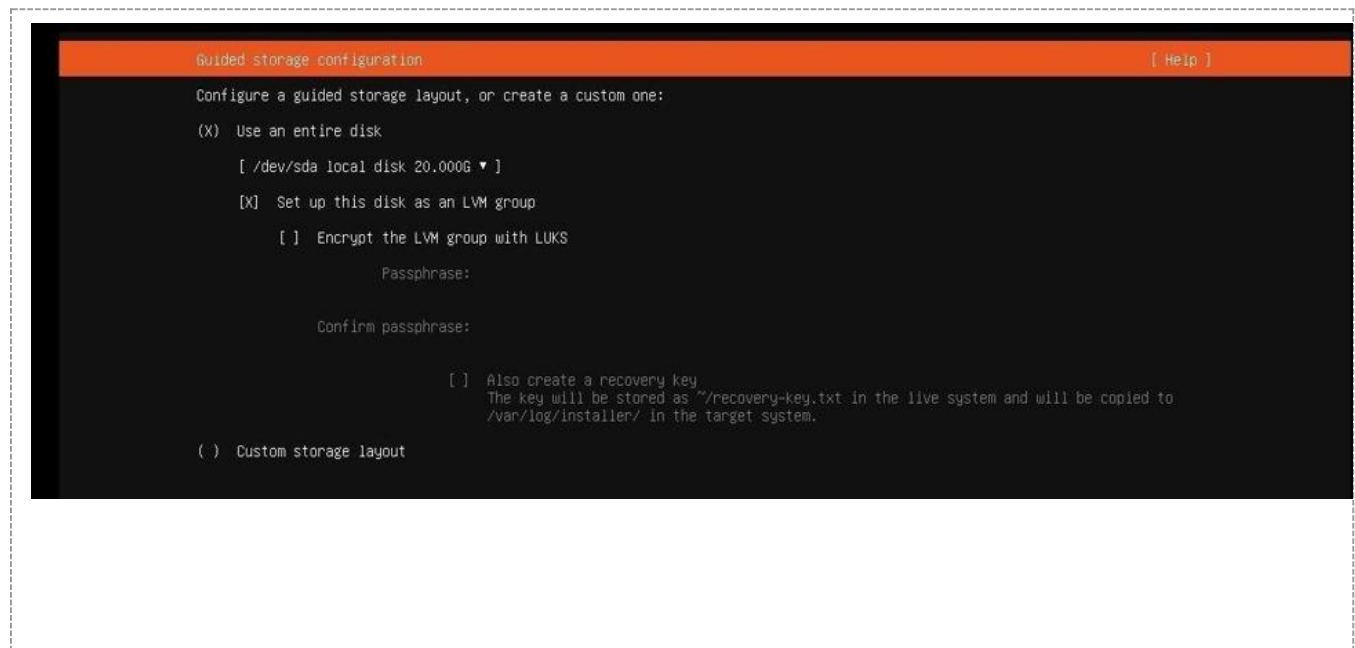
Caption for `ubuntu_installer_network.png`: Automatic IPv4 address assignment received via DHCP.

Task 9: Configure Storage

Select the guided storage option to use the VMware virtual disk.

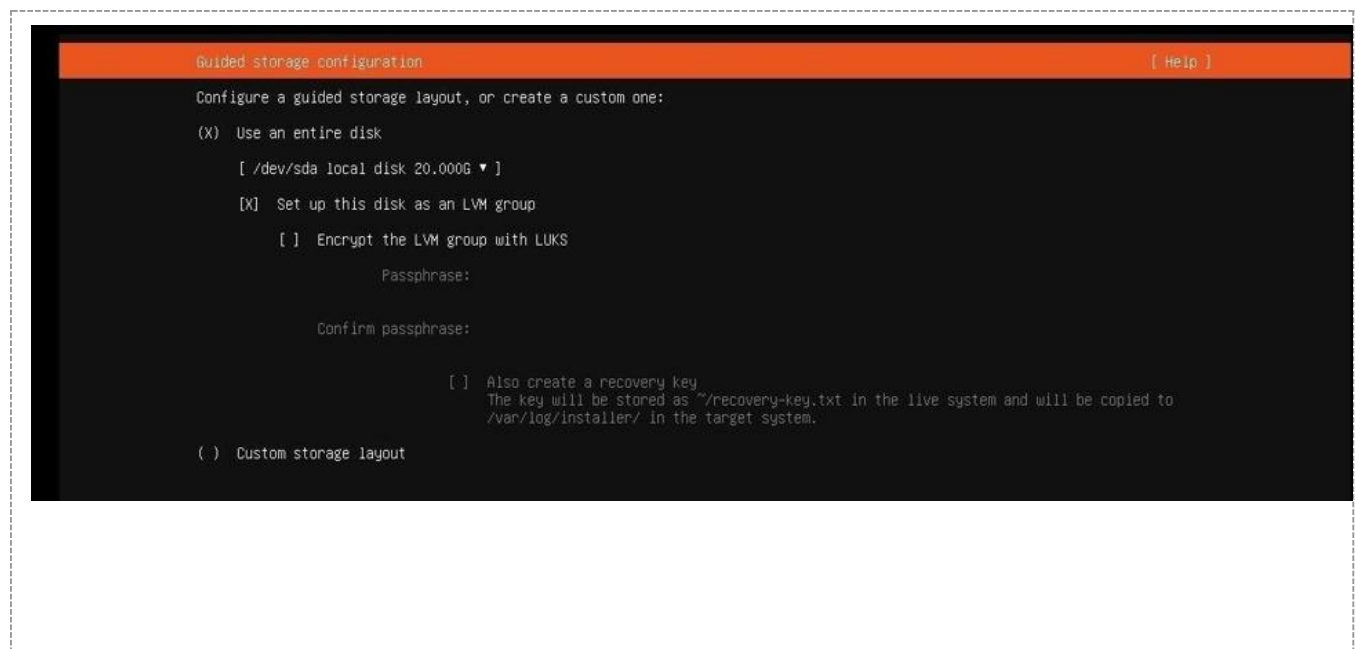
Review the storage configuration before continuing (use only the VMware virtual disk for this lab).

ubuntu_guided_storage.png



Caption for ubuntu_guided_storage.png: Choosing Guided Storage configuration to use the entire virtual disk.

ubuntu_storage_configuration.png



Caption for ubuntu_storage_configuration.png: Summary view of the proposed automated storage layout.

Task 10: Select the Installation Disk

Select the virtual disk created for the Ubuntu Server VM.

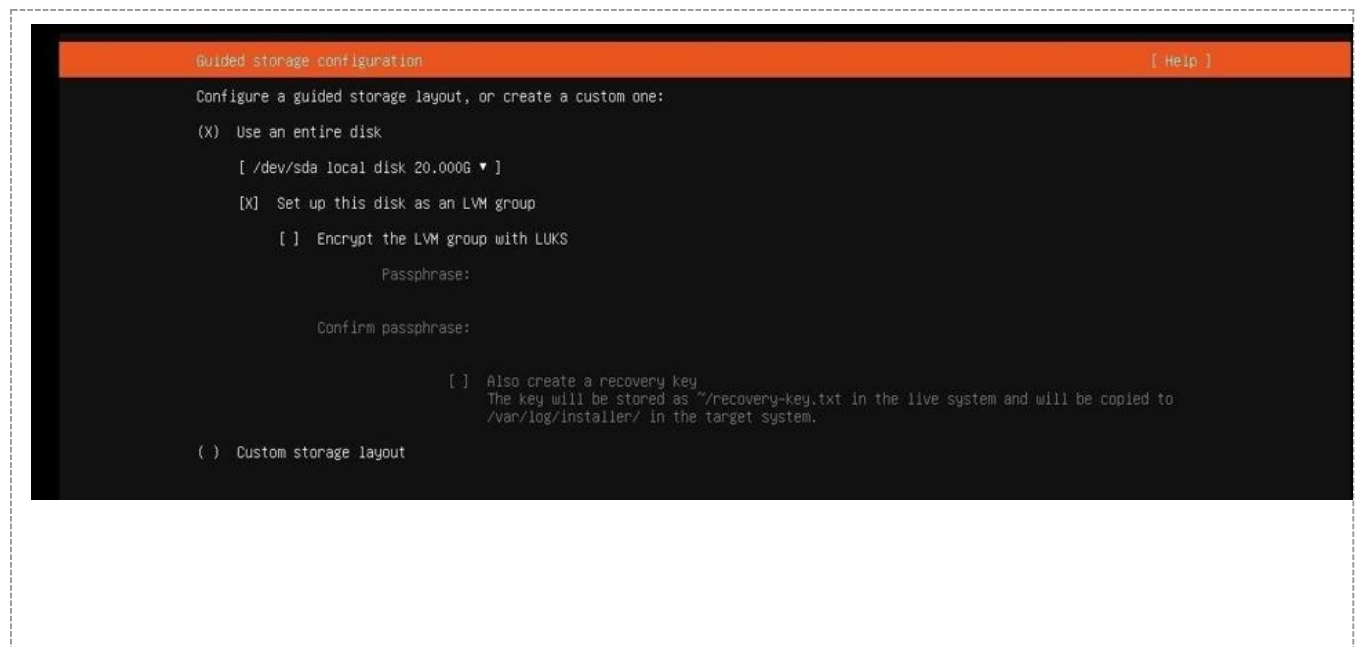
Check the displayed disk name and size to confirm it is the correct VMware virtual disk.

ubuntu_selected_disk.png



Caption for ubuntu_selected_disk.png: Target VMware virtual disk selected for OS installation.

ubuntu_disk_details.png

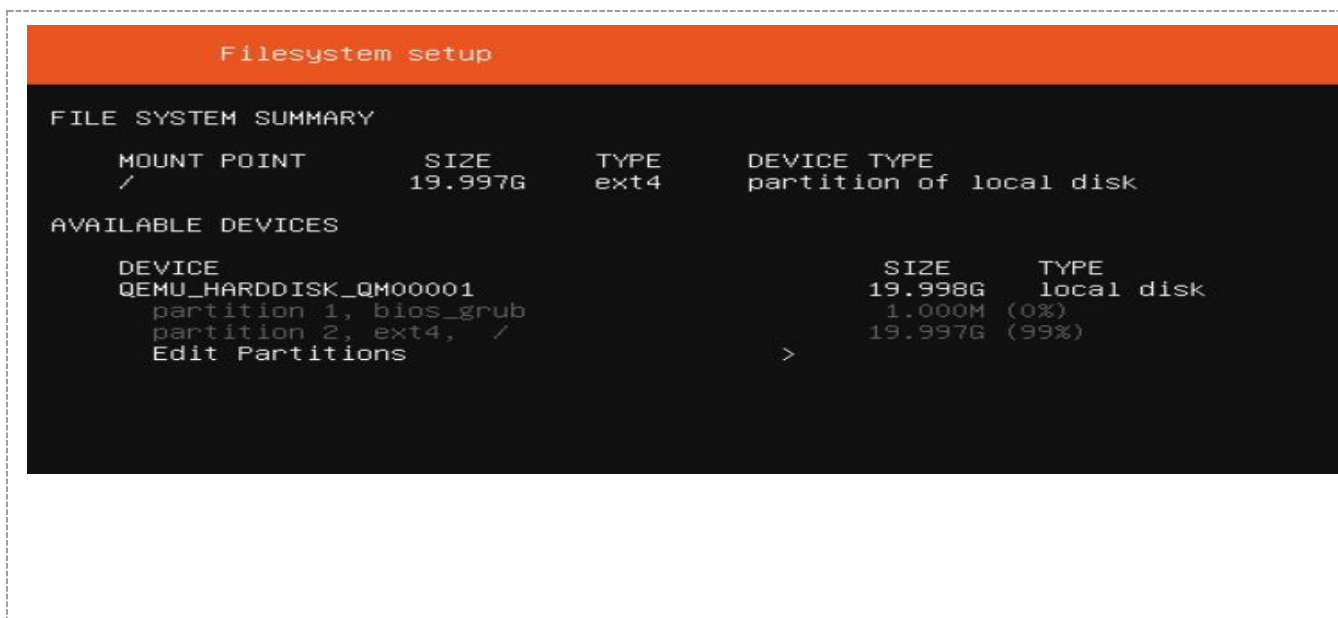


Caption for ubuntu_disk_details.png: Confirming the correct virtual disk capacity and device path.

Task 11: Review the Partitions

Review the partition layout proposed by the installer and confirm the partitions belong to the VMware virtual disk before selecting Done.

ubuntu_partition_layout.png



Caption for ubuntu_partition_layout.png: File system partitioning layout summary for the virtual drive.

Task 12: Confirm the Storage Changes

Read the destructive-action warning carefully.

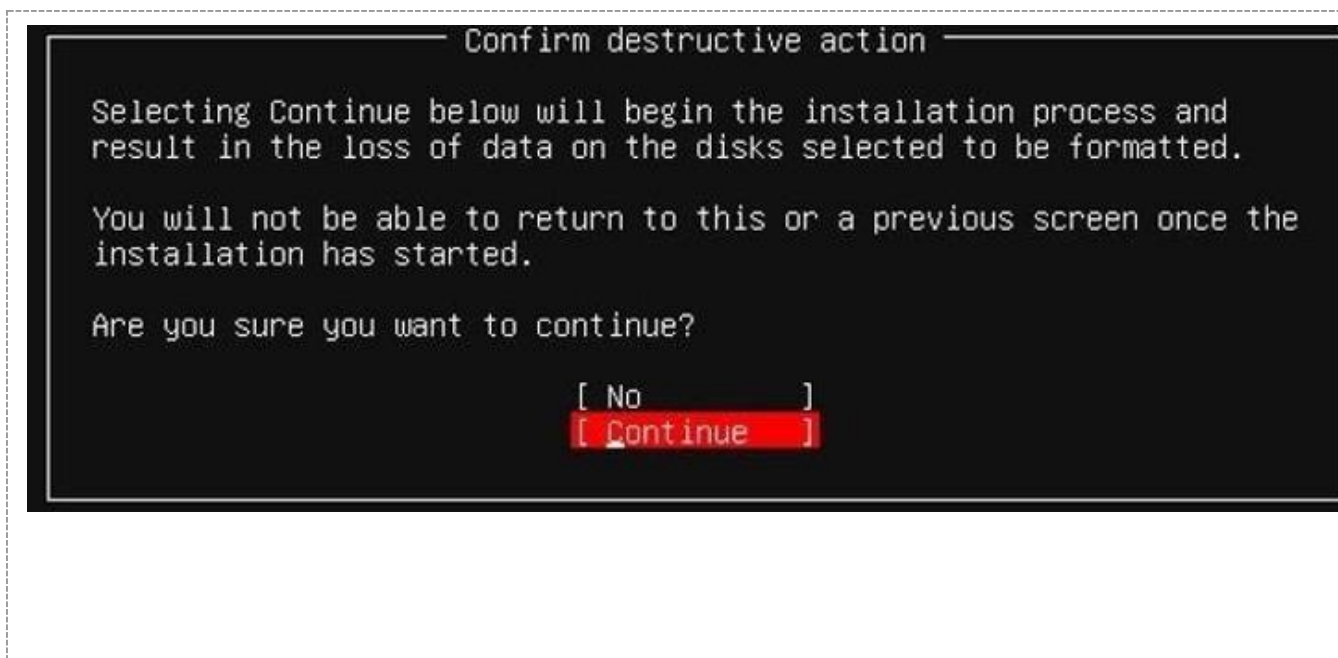
Confirm the changes only when certain the VMware virtual disk is selected.

ubuntu_storage_warning.png



Caption for ubuntu_storage_warning.png: Storage destruction warning prompt prior to formatting.

ubuntu_storage_confirmed.png



Caption for `ubuntu_storage_confirmed.png`: Confirmation of disk changes to proceed with installation.

Task 13: Configure the Ubuntu Profile

Your name: enter your complete actual name.

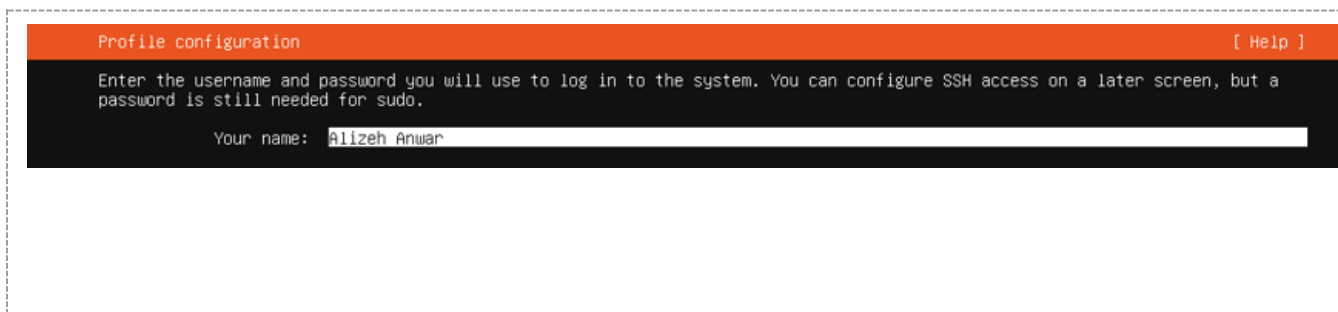
Your server's name: enter 'ubuntu'.

Pick a username: enter your exact GitHub username.

Choose and confirm a secure password (do not expose it in any screenshot).

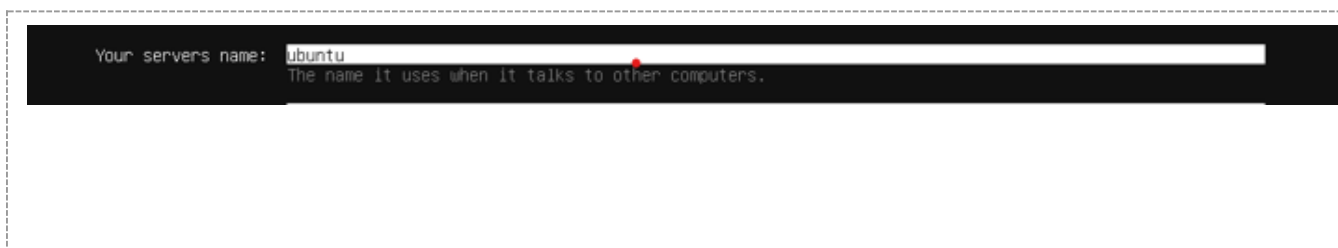
Review all profile fields before continuing.

`ubuntu_actual_name.png`



Caption for `ubuntu_actual_name.png`: Profile setup displaying full actual name "Alizeh Anwar".

`ubuntu_server_name.png`



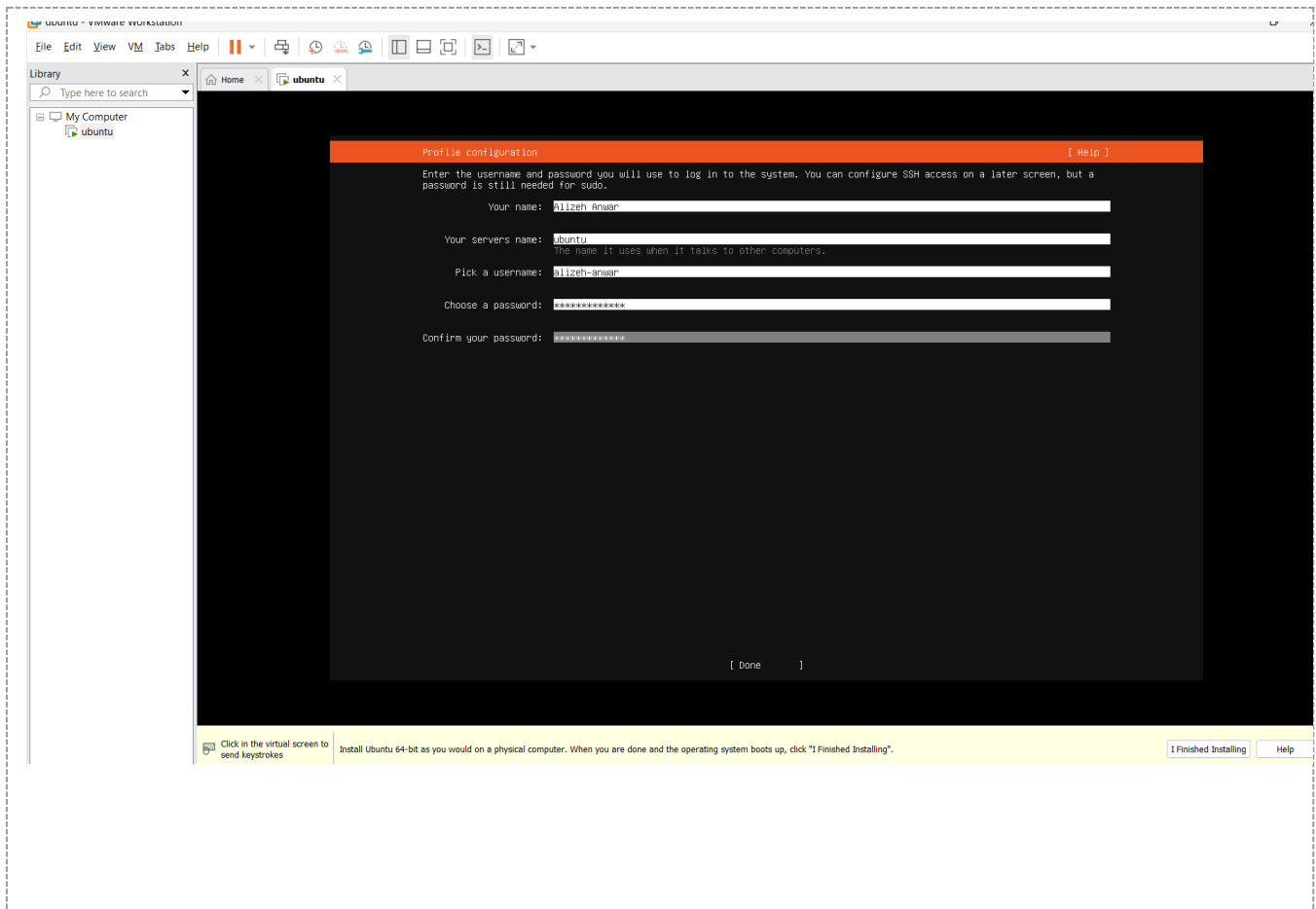
Caption for ubuntu_server_name.png: Server hostname set to ubuntu.

ubuntu_username.png



Caption for ubuntu_username.png: System account username configured as alizabeth-anwar.

ubuntu_profile_setup.png



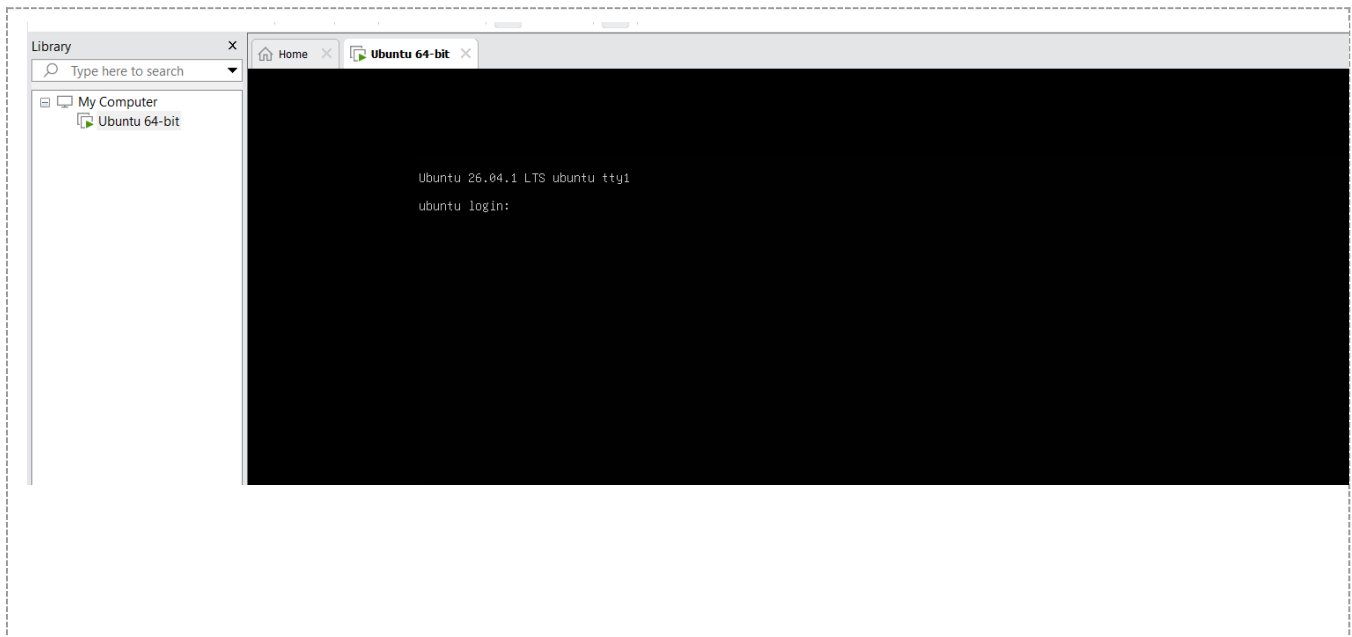
Caption for ubuntu_profile_setup.png: Completed system identity and authentication details.

Task 14: Complete the Software Installation

Wait while Ubuntu Server installs the required software. Do not power off the VM during installation.

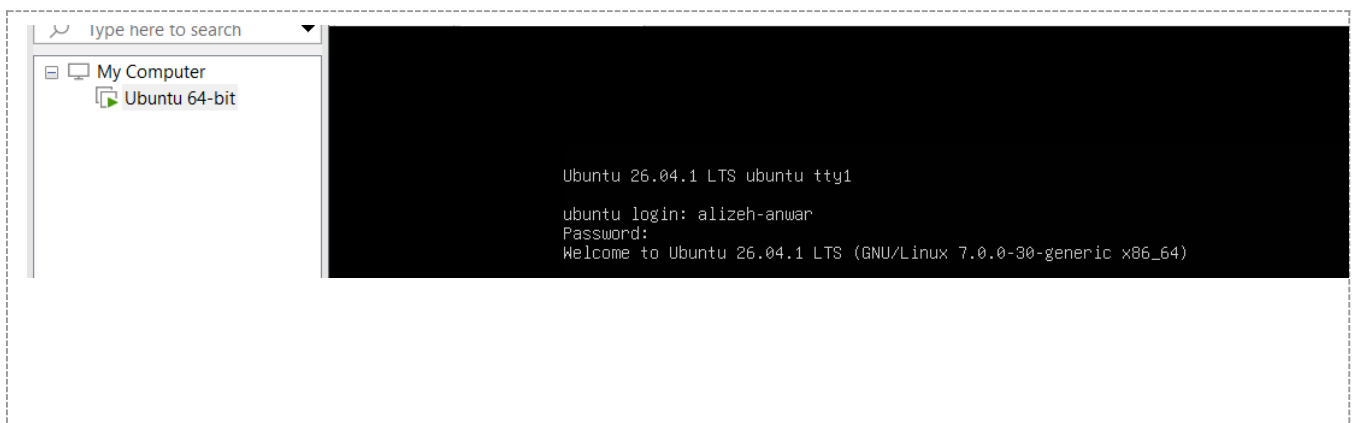
ubuntu_installation_progress.png

ubuntu_login_screen.png



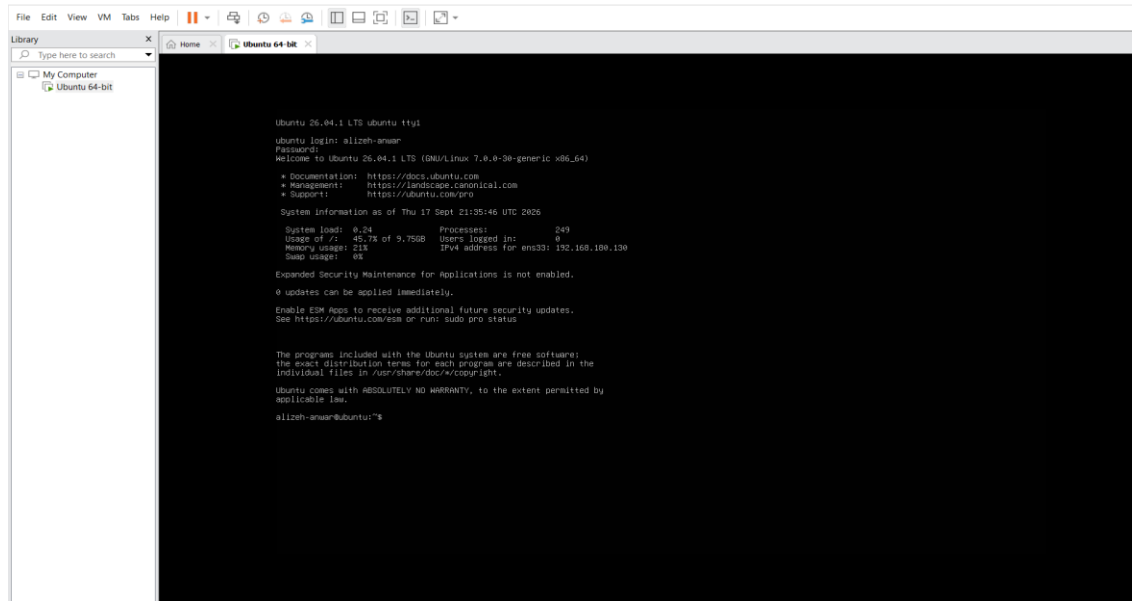
Caption for ubuntu_login_screen.png: Initial login screen after server reboot.

ubuntu_terminal_login.png



Caption for ubuntu_terminal_login.png: Authenticating into local terminal session.

ubuntu_identity_verified.png



Caption for `ubuntu_identity_verified.png`: Local terminal prompt successfully verifying identity as `alizerh-anwar@ubuntu:~$`.

Task 16: Find the Ubuntu Server IP Address

Run: `ip addr`

Find the IPv4 address shown after 'inet' for the active network interface (not 127.0.0.1). Keep the <github-username>@ubuntu prompt visible.

`ubuntu_ip_addr_command.png`

```
Hello Cloud
alizerh-anwar@ubuntu:~$ ip addr
1: lo: <LOOPBACK> UP (LOOPBACK) mtu 65536 qdisc noop state DOWN link-netns default txqueuelen 1000
```

Caption for `ubuntu_ip_addr_command.png`: Running the `ip addr` command in Ubuntu terminal.

`ubuntu_ip_address.png`

```
Hello Cloud
alizeh-anwar@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:83:bc:26 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2983bc26
    inet 192.168.180.130/24 metric 100 brd 192.168.180.255 scope global dynamic ens33
        valid_lft 1089sec preferred_lft 1089sec
    inet6 fe80::20c:29ff:fe83:bc26/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
alizeh-anwar@ubuntu:~$
```

Caption for ubuntu_ip_address.png: Output displaying active network interface IPv4 address.

Task 17: Connect to Ubuntu Server from Windows

Keep the VM powered on. On Windows, open Command Prompt or PowerShell.

Run: `ssh <github-username>@<ubuntu-server-ip>`

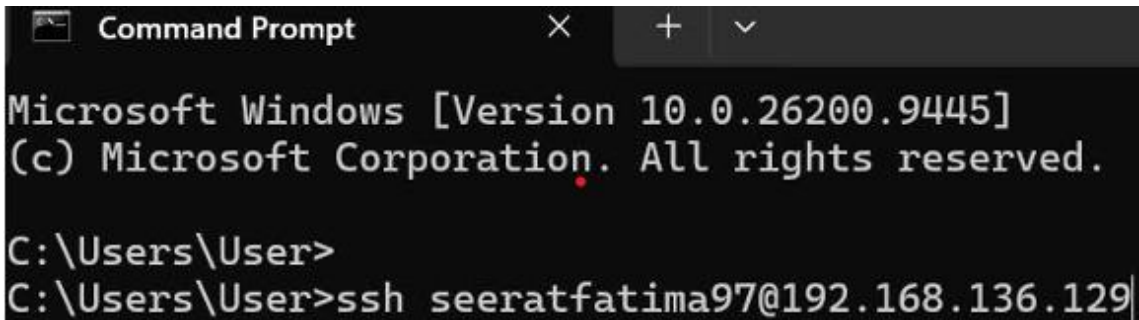
Type 'yes' to accept the server fingerprint if asked.

Enter your Ubuntu password to complete the remote login (do not expose the password in a screenshot).

Verify the remote prompt displays `<github-username>@ubuntu`.

If SSH is unavailable inside Ubuntu, run: `sudo apt update && sudo apt install -y openssh-server && sudo systemctl enable --now ssh`

[windows_ssh_command.png](#)



```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>
C:\Users\User>ssh seeratfatima97@192.168.136.129
```

Caption for windows_ssh_command.png: Windows Command Prompt running `ssh alizeh-anwar@<ubuntu-ip>`.

[windows_ssh_fingerprint.png](#)

```
The authenticity of host '192.168.136.129 (192.168.136.129)' can't be established.  
ED25519 key fingerprint is SHA256:sCrWkqdS70DoCCmjEas2qQzHbIPoNAlaIVfPrPc46Uw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.136.129' (ED25519) to the list of known hosts.
```

Caption for windows_ssh_fingerprint.png: Accepting the SSH host key fingerprint (yes).

windows_ssh_login.png

```
C:\Users\User>ssh seeratfatima97@192.168.136.129  
seeratfatima97@192.168.136.129's password:
```

Caption for windows_ssh_login.png: Remote SSH login completed after authenticating.

windows_ssh_identity.png

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 02:13:04 PM UTC 2026

System load:  0.08               Processes:            219
Usage of /:   50.6% of 9.75GB    Users logged in:     0
Memory usage: 21%               IPv4 address for ens33: 192.168.136.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

seeratfatima97@ubuntu:~$ |
```

Caption for windows_ssh_identity.png: Verified remote Windows terminal prompt displaying alizeh-anwar@ubuntu:~\$.

Task 18: Prepare Lab1 Solution PDF

Add a title page with: Lab number and title, your complete actual name, registration number, course code, section, and GitHub username.

Add headings for Tasks 1 through 17, with screenshots for each task in order and a short caption below each.
Export the completed document as exactly: Lab1_Solution.pdf, and place it at Labs/Lab01/Lab1_Solution.pdf.
Keep all original screenshots inside Labs/Lab01/screenshots/ for automatic grading.

[solution_title_page.png](#)

CLOUD COMPUTING

LAB 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

- **Name:** Alizeh Anwar
- **Reg. no.:** FA24b1-SE-038
- **Course Code:** Cloud Computing (SE-665)
- **Section:** B
- **GitHub username:** alizeh-anwar



Caption for solution_title_page.png: Title page containing student and course details.

Final Submission Checklist

- README.md remains inside Labs/Lab01/.
- The instructor-provided images/install-ubuntu-server/ directory remains unchanged.
- Lab1_Solution.pdf was created by me and placed inside Labs/Lab01/.
- Every original evidence screenshot is inside Labs/Lab01/screenshots/.
- My GitHub profile contains my complete actual name.
- My GitHub username is professional and suitable as an Ubuntu username.
- I used the same GitHub account for the Student Marks Portal.
- I entered my correct registration number, course code, and section.
- Ubuntu Server was installed inside VMware Workstation Pro.
- My Ubuntu username exactly matches my GitHub username.
- My Ubuntu server name is ubuntu.

- My local terminal prompt shows <github-username>@ubuntu.
- I identified the correct Ubuntu Server IPv4 address
- I successfully connected to Ubuntu Server from Windows using SSH.
- My remote SSH prompt shows <github-username>@ubuntu.
- Every screenshot uses the exact required filename.